Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 1 / 方法 1 [LO02-M01]

A discrete optimum is a bound plus a witness. Use integrality, exchange moves, residue capacities and resource counts to narrow the search, then construct an arrangement that meets every local condition.

离散最优值由界与见证共同组成。用整数性、交换操作、余数容量及资源总量缩小搜索，再构造满足全部局部条件的配置。

Bound digit positions, then optimize lexicographically

先界定数位再按字典序优化

Procedure

1. Write every arithmetic digit condition as a weighted place-value equation.

2. Bound the total contribution of lower positions to rule out impossible high positions.

3. Among the remaining strings, optimize the leading digit first while preserving feasible remaining digit mass.

4. Use exchange of digit mass toward cheaper positions to prove a minimum, or the reverse for a maximum.

通用步骤

1. 把数字算术条件写成位权方程。

2. 用低位总贡献的界排除过高数位。

3. 在剩余字符串中先优化首位，同时保留余下数字总量的可行性。

4. 通过把数字总量移向低成本位置的交换证明最小，求最大则反向。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 1 / 方法 1 [LO2-M01] (continued / 续)

> Example. Find the smallest four-digit decimal number whose digits are at most four and sum to ten.

> 例题：求各位数字均不超过四、数字和为十的最小四位十进制数。

Worked solution

1. The leading digit is at least one; choosing one is feasible because the other positions can hold up to twelve.

2. After choosing one, the last two positions hold at most eight, so the hundreds digit must be at least one. Choose it as one.

3. The remaining mass eight forces the final digits four and four. The number is 1144; each earlier digit was proved minimal while completion remained possible.

例题解析

1. 首位至少一，取一可行，因为余三位容量十二。

2. 首位取一后，末两位容量仅八，因此百位至少一，取一。

3. 剩余数字和八迫使末两位均四，得到 1144；每个高位均在可完成条件下取了最小值。

A leading-digit choice is valid only if the remaining positions can complete the weighted condition.

只有余下数位能完成位权条件，首位选择才有效。

Practice / 配套练习: AIME-2001-I-Q08 (WB p. 585), AIME-2020-II-Q05 (WB p. 578)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 2 / 方法 2 [LO02-M02]

Turn a forbidden pattern into forced sorted growth

将禁止模式转为被迫的有序增长

Procedure

1. Sort the positive entries and test the closest preceding entries to a proposed largest member.

2. The absence of a local forbidden pattern gives a recurrence lower bound.

3. Iterate the bound from the smallest permitted initial entries.

4. Check that the equality sequence avoids every nonlocal forbidden pattern too.

通用步骤

1. 将正数排序，对某个最大成员先测试最接近的前两项。

2. 局部禁止模式不存在会给出递推下界。

3. 从最小允许初项迭代该界。

4. 检查取等序列也能避免所有非局部禁止模式。

> Example. Prove that a sorted positive list contains a nondegenerate triangle if and only if some three consecutive entries form one.

> 例题：证明递增正数列表含有非退化三角形，当且仅当某三个连续成员可作三角形。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 2 / 方法 2 [LO02-M02] (continued / 续)

Worked solution

1. One direction is immediate: a consecutive triple forming a triangle is also a triple in the whole list.

2. Otherwise every consecutive triple obeys $a_{j-2} + a_{j-1} \leq a_j$.

3. For arbitrary $i < k < j$, $a_i + a_k \leq a_{j-2} + a_{j-1} \leq a_j$, so no arbitrary triple forms a triangle. This proves the converse and the exhaustive local test.

例题解析

1. 连续三项若能作三角形，自然给出所需三项。

2. 否则每个连续三项均有前两项之和不超过第三项。

3. 任意更早两项之和也不超过最大项前的两项之和，故同样不能作三角形，完成反向证明。

A locally defined extremal sequence needs a proof that it also avoids nonconsecutive patterns.

局部定义的极值序列还须证明能避免非连续模式。

Practice / 配套练习: AIME-2001-II-Q05 (WB p. 800)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 3 / 方法 3 [LO02-M03]

Combine a common residue class with a distinctness capacity

结合公共余数类与互异容量

Procedure

1. Translate every deletion or divisibility condition into a common congruence for all entries.

2. Use the difference of the largest and smallest entries to restrict the modulus.

3. Distinct entries in one residue class require minimum successive gaps equal to that modulus.

4. Construct an initial arithmetic block plus the prescribed endpoint and verify every deletion condition.

通用步骤

1. 把每次删除或整除条件化为所有成员的公共同余。

2. 由最大最小差限制模数。

3. 同余类内互异成员的连续间距至少为该模数。

4. 用起始等差块加规定端点构造，并验证每次删除。

> Example. For positive integers $m < M$ and $n \geq 2$, characterize when an n-element set with minimum m, maximum M has integer mean after any one deletion.

> 例题：给定正整数 $m < M, n \geq 2$，刻画何时存在最小值 m、最大值 M 的 n 元集，删除任一成员后的平均数均为整数。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 3 / 方法 3 [LO02-M03] (continued / 续)

Worked solution

1. Put $d = n - 1$ and let the sum be S. Every deletion gives $S - x \equiv 0 \pmod{d}$, so all entries are congruent and $d \mid M - m$.

2. There are d positive gaps of at least d, so $M - m \geq d^2$.

3. Conversely, if both conditions hold, take $m, m + d, \ldots, m + (d - 1)d, M$. They are distinct, and all entries are congruent to m; the total is also $m \pmod{d}$. Thus every deletion sum is divisible by d.

例题解析

1. 令 $d = n - 1$，总和为 S。每次删除给出 $S - x \equiv 0 \pmod{d}$，故所有成员同余，且 $d \mid M - m$。

2. 共有 d 个正间距，每个至少 d，所以 $M - m \geq d^2$。

3. 反之，若两条件都满足，取起始 d 项等差块及端点 M，全部互异且同余；总和也与 m 同余，故任删一项后被 d 整除。

Prove both the divisor restriction and the spacing capacity; either alone is too weak.

约数限制与间距容量两者都要证明，单用一者不够。

Practice / 配套练习: AIME-2002-I-Q14 (WB p. 508)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 4 / 方法 4 [LO02-M04]

Round per-group bounds before summing integer totals

先对各组整数界取整再求和

Procedure

1. Convert percentage or ratio constraints to exact inequalities with positive denominators.

2. For strict upper bounds on integers use $x < r \Rightarrow x \le \lceil r \rceil - 1$; for lower bounds use ceilings.

3. Add the local integer bounds and compare with the global total.

4. Distribute any remaining excess explicitly and verify every group separately.

通用步骤

1. 将百分比或比例限制化为正分母的精确不等式。

2. 整数严格上界用 $x < r \Rightarrow x \le \lceil r \rceil - 1$，下界用向上取整。

3. 相加各局部整数界，再与全局总量比较。

4. 明确分配剩余量，并逐组验证。

> Example. Three bags have capacities $5, 11, 16$. Each contains a positive integer number of marked items, strictly less than five-eighths of its capacity. Maximize the total marked count.

> 例题：三个袋子容量为五、十一、十六，标记物数量均为正整数且严格小于各容量的八分之五，求标记总数最大值。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 4 / 方法 4 [LO02-M04] (continued / 续)

Worked solution

1. The three integer upper bounds are $\lceil 25/8 \rceil - 1 = 3$, $\lceil 55/8 \rceil - 1 = 6$, and $10 - 1 = 9$.

2. Thus at most eighteen items are marked, stronger than rounding only the combined capacity bound.

3. Counts $(3, 6, 9)$ satisfy all strict per-bag inequalities and attain eighteen.

例题解析

1. 三组整数上界分别为三、六、九。

2. 所以总数至多十八，比只对合并容量取整更强。

3. 取标记数三、六、九，逐组均严格符合比例，故十八可达。

A strict ratio bound can lose a full integer when its unrounded bound is itself integral.

当未取整上界本身为整数时，严格比例还须再减一。

Practice / 配套练习: AIME-2003-II-Q12 (WB p. 806), AIME-2004-I-Q05 (WB p. 801)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 5 / 方法 5 [LO02-M05]

Use unit exchanges to prove no gaps in attainable sums

用单位交换证明可达和无缺口

Procedure

1. Compute the smallest and largest possible sums.

2. At every nonmaximal configuration find a legal exchange increasing the sum by exactly one.

3. The sum increases within a finite set, so the exchanges eventually reach the maximum.

4. Every intermediate integer has then been attained; count the full interval.

通用步骤

1. 先算最小及最大可能总和。

2. 在每个非最大配置中，找出合法且恰增加一的交换。

3. 总和在有限集合中严格增加，故最终到达最大值。

4. 于是每个中间整数都达到，可直接数完整区间。

> Example. Prove the sums of k-element subsets of $\{1, \ldots, N\}$ fill an integer interval, and find its size.

> 例题：证明一至 N 的 k 元子集和填满一个整数区间，并求区间大小。

Worked solution

1. The endpoint sums are $k(k+1)/2$ and $k(2N-k+1)/2$.

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离散最值与构造

Method 5 / 方法 5 [LO02-M05] (continued / 续)

2. Unless the set is the terminal block, some chosen $j < N$ has unchosen successor $j + 1$. Replacing it legally raises the sum by one.

3. Following such exchanges from the initial block reaches the terminal block and visits every integer between. The number of sums is $k(N - k) + 1$.

例题解析

1. 两端和分别为最小前块及最大末块之和。

2. 若还不是末块，必有被选 j 的后继 $j + 1$ 未选，替换后合法且和增加一。

3. 从前块不断交换到末块，中间整数无遗漏，故总数为 $k(N - k) + 1$。

Knowing the endpoint sums is insufficient without the no-gap exchange proof.

只知道两端和还不够，必须证明中间无缺口。

Practice / 配套练习: AIME-2006-I-Q02 (WB p. 802)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 6 / 方法 6 [LO02-M06]

Telescope to a quantized terminal state and construct it

望远镜化到离散终态并构造

Procedure

1. Square an absolute recurrence to remove the arbitrary sign.

2. Sum the resulting differences so the objective depends only on a terminal square.

3. Prove which terminal parity and size classes are attainable.

4. Choose the best allowed terminal value and give a complete legal sign sequence reaching it.

通用步骤

1. 将绝对值递推平方，消去任意符号。

2. 累加差式，使目标只依赖终态平方。

3. 证明终态允许的奇偶及大小类。

4. 选择最优允许终值，并给出完整合法符号序列。

> Example. Let $u_0 = 0, |u_j| = |u_{j-1} + 1|$. Find all possible values of $u_0 + u_1 + u_2 + u_3$.

> 例题：设 $u_0 = 0, |u_j| = |u_{j-1} + 1|$，求前四项和的全部可能值。

Worked solution

1. Squaring and summing through step four gives $u_4^2 = 4 + 2\sum_{j=0}^{3} u_j$.

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离散最值与构造

Method 6 / 方法 6 [LO02-M06] (continued / 续)

2. Induction on the recurrence gives terminal values $u_4 \in \{-4, -2, 0, 2, 4\}$, all attained: each stage gives every integer of its parity between its two extreme values.

3. The terminal squares are $16, 4, 0$, so the sums are exactly $6, 0, -2$. The reachability induction supplies every listed value.

例题解析

1. 平方后累加到第四步，得 $u_4^2 = 4 + 2\sum_{j=0}^{3} u_j$。

2. 递推归纳得第四步可达负四、负二、零、二、四：每步均填满其两端之间的同奇偶整数。

3. 终态平方为十六、四、零，故和恰为六、零、负二，每个值均由可达性归纳保证。

A nearest-square bound becomes an optimum only when the terminal state can actually occur.

最近平方给出的界，只有终态确实可达时才成为最优值。

Practice / 配套练习: AIME-2006-I-Q15 (WB p. 156)

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离散最值与构造

Method 7 / 方法 7 [LO02-M07]

Check the nearest admissible integers of a concave product

比较凹乘积顶点附近的允许整数

Procedure

1. Eliminate one count using the linear resource equation.

2. Complete the square in the remaining concave quadratic.

3. Solve its integrality congruence and test the nearest admissible points on both sides of the vertex.

4. Convert the maximizing counts back to a valid geometric or combinatorial arrangement.

通用步骤

1. 用线性资源方程消去一个计数。

2. 对剩下的凹二次式配方。

3. 解整数性同余，并检查顶点两侧最近的允许点。

4. 将最优计数恢复为合法几何或组合配置。

> Example. Positive integers $x+y=12$ satisfy $x \equiv 1 \pmod 3$. Maximize xy.

> 例题：正整数 $x+y=12$ 且 $x \equiv 1 \pmod 3$，求 xy 最大值。

Worked solution

1. The product is $36-(x-6)^2$.

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离散最值与构造

Method 7 / 方法 7 [LO02-M07] (continued / 续)

2. Allowed x are $1, 4, 7, 10$. The closest to six is seven, at distance one; four is distance two.

3. Thus $(x, y) = (7, 5)$ attains the maximum thirty-five.

例题解析

1. 乘积可配方为 $36 - (x - 6)^2$。

2. 允许的一、四、七、十中，七距顶点最近。

3. 取七、五得最大乘积三十五。

Rounding the real vertex without enforcing the residue class can give an impossible count.

若不执行余数限制就将实顶点取整，可能得到不可能的计数。

Practice / 配套练习: AIME-2007-II-Q08 (WB p. 803)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 8 / 方法 8 [LO02-M08]

Distinguish a finite-window count from a steady rate

区分有限时间窗计数与稳定速率

Procedure

1. Translate the spacing rule to a lower bound on time between successive events.

2. m events in a closed window require only $m-1$ inter-event gaps.

3. Use strict gap inequalities to rule out the next integer count.

4. Choose an initial phase and explicit legal gaps that attain the claimed count.

通用步骤

1. 将间距规则转为连续事件之间的时间下界。

2. 闭时间窗内的 m 个事件只需 $m-1$ 个间隔。

3. 用严格间隔不等式排除下一个整数计数。

4. 选择初始相位及明确合法间隔，达到所称数量。

> Example. Events are separated by strictly more than two minutes. What is the largest number in the closed ten-minute interval $[0,10]$?

> 例题：相邻事件间隔严格大于两分钟，在闭区间零至十分钟内最多可有几个?

Worked solution

1. Six events would require five gaps whose total is strictly greater than ten, impossible.

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离散最值与构造

Method 8 / 方法 8 [LO02-M08] (continued / 续)

2. Five events at times $0, 9/4, 9/2, 27/4, 9$ have gaps $9/4 > 2$ and fit in the window.

3. Thus five is the maximum. An initial event is counted even though it does not require an earlier gap inside the window.

例题解析

1. 六个事件需五个间隔，总长严格大于十，矛盾。

2. 取 $0, 9/4, 9/2, 27/4, 9$ 分钟这五个时刻，间隔均为 $9/4 > 2$。

3. 因此最多五个，初始事件不需要窗内更早的一个间隔。

State whether endpoints and a freely chosen initial phase are included before converting flow to a count.

把流率变成数量前，先说明端点及自由初相是否计入。

Practice / 配套练习: AIME-2008-I-Q12 (WB p. 173)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 9 / 方法 9 [LO02-M09]

Bound a growing path by its distinct distance levels

用不同距离层数限制递增长路径

Procedure

1. List every possible positive pair distance or squared distance.

2. A strictly increasing path uses each level at most once, giving a point-count bound of one more than the number of levels.

3. If the bound is reached, every level must be used in order; backtrack from the largest level to force predecessors.

4. Check distinct vertices and construct or count all surviving paths.

通用步骤

1. 列出所有可能正距离或距离平方。

2. 严格递增路径每层至多一次，故顶点数至多距离层数加一。

3. 若取等，必须按顺序使用全部距离层，可从最大层倒推前驱。

4. 检查顶点互异，构造或计数全部剩余路径。

> Example. Among the four vertices of a 1×2 rectangle, what is the longest path visiting distinct vertices with strictly increasing successive distances?

> 例题：在一乘二矩形的四个顶点中，以不同顶点构成路径且相邻距离严格增加，最多经过几个点?

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 9 / 方法 9 [LO02-M09] (continued / 续)

Worked solution

1. The distance levels are $1, 2, \sqrt{5}$, suggesting an upper bound of four points.

2. A four-point path would need distances $1, 2, \sqrt{5}$ in order. The first two are perpendicular rectangle sides, after which the unused fourth vertex is distance one from the current vertex, not $\sqrt{5}$. Thus four is impossible.

3. Three points are possible by a length-one side followed by a diagonal. Hence the true maximum is three; the level bound alone was not attained.

例题解析

1. 三种距离是一、二、根号五，形式上顶点数至多四。

2. 若走四点，距离须依次一、二、根号五；前两段为相邻垂直边，此时唯一未用顶点距当前位置却是一，不能完成。

3. 先走短边再走对角线可走三点，所以准确最大为三，距离层数界并不自动可达。

A level-count upper bound is only a candidate until a path realizes its entire ordering.

只有有路径实现完整距离顺序，层数界才可达到。

Practice / 配套练习: AIME-2008-II-Q10 (WB p. 697)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 10 / 方法 10 [LO02-M10]

Count one resource in two ways and build a compatible block

从两种角度计同一资源并构造相容块

Procedure

1. Write the same total as a sum over each of two families of groups.

2. Use distinctness or fixed per-group counts to give lower and upper bounds.

3. Solve the resulting divisibility or size obstruction.

4. Build repeated local blocks or disjoint parity classes to satisfy every local rule, not only the total equation.

通用步骤

1. 按两类分组分别表示同一个总量。

2. 由互异性或固定组内数量给出上下界。

3. 解所得整除或大小障碍。

4. 用重复局部块或不相交奇偶类构造，使全部局部规则成立，而不只满足总量方程。

> Example. A nonempty zero-one matrix has two ones in every row and three ones in every column. Find the least possible number of entries.

> 例题：非空零一矩阵每行两个一、每列三个一，求总格数最小值。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 10 / 方法 10 [LO02-M10] (continued / 续)

Worked solution

1. With r rows and c columns, counting ones gives $2r = 3c$, hence $r = 3t, c = 2t$ for positive integer t.

2. Therefore the entry count is $6t^2 \geq 6$.

3. The 3×2 all-one matrix meets both local rules and attains six entries, proving the minimum.

例题解析

1. 按行列计一的总数，得 $2r = 3c$，所以行列数为 $3t, 2t$。

2. 总格数为 $6t^2 \geq 6$。

3. 三乘二全一矩阵满足各行各列要求并达到六，故最小值为六。

Equality of global counts does not guarantee a compatible local arrangement; exhibit one.

全局数量相等不保证局部配置相容，必须给出见证。

For k disjoint pairs from $\{1, \ldots, N\}$, the used-label sum is at least $k(2k+1)$. If pair sums are distinct and at most N, their sum is at most $k(2N-k+1)/2$. Comparing the same total gives $5k \leq 2N-1$. A sharp construction must separately ensure disjoint labels and distinct sums, often by giving one block even sums and another odd sums.

从一至 N 选 k 个不交元素对，已用标号总和至少 $k(2k+1)$。若对和互异且不超过 N，其总和至多 $k(2N-k+1)/2$，比较同一总量得 $5k \leq 2N-1$。取等构造还须分别保证标号不重、对和互异，常可用一块给偶和、另一块给奇和。

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离散最值与构造

Method 10 / 方法 10 [LO02-M10] (continued / 续)

Practice / 配套练习: AIME-2009-II-Q12 (WB p. 807), AIME-2016-II-Q04 (WB p. 659)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 11 / 方法 11 [LO02-M11]

Build product-free color bands with endpoint checks

以端点核查构造无同色积色带

Procedure

1. Choose multiplicative scale thresholds so products of the low band jump to the other color.

2. Put a far high band in the first color only when multiplying it by any allowed factor leaves the whole domain.

3. Check products within the other color land outside that color.

4. Include every endpoint and permit repeated factors when the original relation allows them.

通用步骤

1. 选择乘法尺度门槛，使低色带乘积跳到另一颜色。

2. 只有高色带乘任一允许因子都超出全域时，才可把它放回第一色。

3. 检查另一色内部的积不会仍落在本色。

4. 逐一核查端点，并保留原关系允许的重复因子。

> Example. For any integer $a \geq 2$, construct a two-coloring of $\{a, \ldots, a^5 - 1\}$ with no monochromatic product $xy = z$.

> 例题：对任意整数 $a \geq 2$，构造从 a 至 $a^5 - 1$ 的两色染色，使无同色乘法关系。

Worked solution

1. Color $[a, a^2 - 1] \cup [a^4, a^5 - 1]$ red and $[a^2, a^4 - 1]$ blue.

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离散最值与构造

Method 11 / 方法 11 [LO02-M11] (continued / 续)

2. Two low red factors multiply into the blue band; a product involving a high red factor is at least a^5 and leaves the domain.

3. Two blue factors have product at least a^4, so any in-domain product is red. Thus every same-color product is excluded, including equal factors.

例题解析

1. 将整数带 $[a, a^2-1] \cup [a^4, a^5-1]$ 染红，将 $[a^2, a^4-1]$ 染蓝。

2. 两低红因子的积落在蓝带；含高红因子的积至少 a^5，已出域。

3. 两蓝因子的积至少 a^4，若仍在域内则为红，因此同色积关系全部排除，包括重复因子。

The high band is safe only because its product with the smallest allowed factor is already outside the domain.

高色带安全的关键，是它乘最小允许因子也已越界。

Practice / 配套练习: AIME-2010-I-Q12 (WB p. 798)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 12 / 方法 12 [LO02-M12]

Saturate a run-count bound before counting compositions

先使连续段界取满再数正整数组成

Procedure

1. Bound the number of color changes or runs by the scarcer symbol count.

2. Translate any equal-pair condition into a bound on total adjacent pairs.

3. At equality, identify all forced singleton runs and endpoint colors.

4. Count the remaining positive run lengths by compositions, restoring labels only if the objects are distinct.

通用步骤

1. 用稀少符号数量限制颜色变化数或连续段数。

2. 将相邻对数量关系转为总相邻对界。

3. 取等时找出所有被迫的单元素段及端点颜色。

4. 用正整数组成数剩余段长，只有对象不同才恢复标号。

> Example. A binary word has three ones and seven zeros. Maximize its color changes and count the maximizing words.

> 例题：一个二进制词有三个一、七个零，求最多变色次数及达到最大值的词数。

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离散最值与构造

Method 12 / 方法 12 [LO02-M12] (continued / 续)

Worked solution

1. Three one-runs allow at most six changes. To reach six, all ones are singletons and there are four positive zero-runs, including both endpoints.

2. Write their lengths as $z_1 + z_2 + z_3 + z_4 = 7$ with each positive.

3. There are $\binom{6}{3} = 20$ such compositions, each giving exactly six changes. Thus the maximum is six and the count twenty.

例题解析

1. 三个一段至多产生六次变色，取等须三个一均单独，且四个零段均非空，包含两端。

2. 四个正零段长之和为七。

3. 隔板法得 $\binom{6}{3} = 20$ 种，每种均恰有六次变色，因此最大次数为六，取等词数为二十。

Counting before analyzing equality can include run patterns that do not attain the bound.

尚未分析取等就计数，会混入未达到上界的段型。

Practice / 配套练习: AIME-2011-II-Q07 (WB p. 652)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 13 / 方法 13 [LO02-M13]

Optimize uncoupled choices term by term

逐项优化互不耦合的选择

Procedure

1. Express the total objective as a sum of contributions, one from each choice.

2. Prove that choosing one option imposes no restriction on the others.

3. Take the maximum contribution independently for each term.

4. Give the actual full selection attaining all local maxima simultaneously.

通用步骤

1. 将总目标拆为每次选择的一项贡献。

2. 证明一个选项不限制其他选择。

3. 每项独立取最大贡献。

4. 给出同时实现全部局部最大值的完整选择。

> Example. For each of $1-2i, -3+4i, 5-i$, choose the number or its conjugate. Maximize the imaginary part of their sum.

> 例题：对 $1-2i, -3+4i, 5-i$ 各选原数或共轭，求总和虚部最大值。

Worked solution

1. Each conjugation can change only the sign of its own imaginary part.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 13 / 方法 13 [LO02-M13] (continued / 续)

2. The independent maxima are two, four, one, summing to seven.

3. Choose $1+2i, -3+4i, 5+i$. Their sum is $3+7i$, attaining the bound.

例题解析

1. 每次共轭只改变自身虚部符号。

2. 三个独立最大虚部分别为二、四、一，总和七。

3. 选取 $1+2i, -3+4i, 5+i$，总和为 $3+7i$，确取等。

Local maximization fails when a shared constraint couples the choices.

若存在耦合选择的共同限制，局部最优不一定能同时实现。

Practice / 配套练习: AIME-2011-II-Q08 (WB p. 72)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 14 / 方法 14 [LO02-M14]

Find a first omitted block using consecutive square gaps

用连续平方间隙寻找首个空块

Procedure

1. Use square gaps $2j+1$ to exclude all blocks too short to fit between consecutive squares.

2. For the remaining aligned blocks, track the quotient $\lfloor j^2/B \rfloor$.

3. A missing block occurs when consecutive square quotients skip its index.

4. Exhibit the two bounding consecutive squares and prove every earlier block had a square.

通用步骤

1. 用平方间距排除无法完整藏在相邻平方间的早期块。

2. 对剩下的对齐块，跟踪平方除以块长的整数商。

3. 当相邻平方的块下标跳过某个整数，就出现空块。

4. 给出夹住该块的连续平方，并证明此前各块都含平方。

> Example. For positive integer B and nonnegative integer N, give an exact test for whether the integer block $BN, \ldots, BN+B-1$ contains a square.

> 例题：设 B 为正整数、N 为非负整数，给出整数块 BN 至 $BN+B-1$ 是否含平方数的精确判断式。

Worked solution

1. The least square at least BN is q^2 with $q=\lceil \sqrt{BN} \rceil$.

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离散最值与构造

Method 14 / 方法 14 [LO02-M14] (continued / 续)

2. If $q^2 \le BN + B - 1$, that square belongs to the block.

3. If it is larger, every subsequent square is even larger and no earlier square reaches BN. Thus the block is empty exactly when $\lceil \sqrt{BN} \rceil^2 > BN + B - 1$.

例题解析

1. 不小于块起点的最小平方为 q^2，其中 $q = \lceil \sqrt{BN} \rceil$。

2. 若它不超过块终点，则块内有该平方。

3. 否则更大平方都越界、更小平方都不及起点，因此空块的充要条件正是该平方严格超过终点。

A large square gap alone does not guarantee that an aligned whole block fits inside it.

平方间距大，并不保证某个对齐整块恰好放在其中。

Practice / 配套练习: AIME-2013-II-Q06 (WB p. 548)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 15 / 方法 15 [LO02-M15]

Use competing integer bounds to force a small candidate class

用相竞争的整数界迫使小候选类

Procedure

1. Find one feasible candidate to establish a numerical benchmark.

2. Use a global denominator or resource bound to exclude all objectives with a larger integer numerator or quotient class.

3. Apply the remaining local bound to enumerate only potentially improving cases.

4. Check every surviving case and retain the original witness if none improves it.

通用步骤

1. 先找一个可行候选作为数值基准。

2. 用全局分母或资源界，排除整数分子或商类别较大的情况。

3. 用另一局部界，只枚举可能改进的少量情形。

4. 检查所有保留候选，若无改进则原见证最优。

> Example. Positive integers a, b satisfy $b \leq 30$ and $a/b < 1/17$. Determine their possible form.

> 例题：正整数 a, b 满足 $b \leq 30$ 且 $a/b < 1/17$，求它们的全部可能形式。

Worked solution

1. If $a \geq 2$, then $a/b \geq 2/30 = 1/15 > 1/17$, impossible.

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离散最值与构造

Method 15 / 方法 15 [LO02-M15] (continued / 续)

2. Thus $a = 1$, and the strict comparison gives $b > 17$.

3. The possibilities are exactly $(a, b) = (1, b)$ for $18 \le b \le 30$, each of which satisfies both conditions.

例题解析

1. 若分子至少二，分数至少十五分之一，比十七分之一还大，矛盾。

2. 所以分子为一，严格比较要求分母大于十七。

3. 因此恰为分子一、分母十八至三十，且全部满足条件。

A good candidate converts a huge search into the small class capable of beating it.

一个好候选能把巨大搜索缩成仅有可能胜过它的小类。

For a remainder of n by $k \le n/2$, the quotient is at least two, so the remainder is bounded by both $k - 1$ and $n - 2k$. Split the divisor range near their intersection, then exhibit a divisor realizing the better bound and check its actual quotient; a divisor of n has remainder zero regardless of a loose upper bound.

用 $k \le n/2$ 除 n 时商至少二，所以余数同时不超过 $k - 1$ 与 $n - 2k$。在两界交会附近分割除数区间，再给出达到较好界的除数并检查真实商；整除时余数为零，不能把宽松上界当作实际余数。

Practice / 配套练习: AIME-2013-II-Q14 (WB p. 514), AIME-2022-I-Q07 (WB p. 804)

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离散最值与构造

Method 16 / 方法 16 [LO02-M16]

Use least representable residues to certify a largest gap

用最小可表示余数值核实最大缺口

Procedure

1. First check that the denominations have gcd one; otherwise infinitely many amounts are missing. Choose a denomination d and find the least representable amount M_r in every residue class modulo d.

2. Adding copies of d then reaches every larger amount in that residue.

3. The last possible gap in residue r is $M_r - d$; maximize these values. If that maximum is negative, no nonnegative amount is missing.

4. Prove every residue minimum is minimal and construct its representation.

通用步骤

1. 先检验所有面额的最大公因数为一，否则有无穷多个金额无法表示。选一种面额 d 作模数，在每个余数类找最小可表示金额 M_r。

2. 再加该面额，就能到达同余类中全部更大金额。

3. 余数类 r 的最后可能缺口为 $M_r - d$，再对各类取最大；若最大值仍为负，则所有非负金额都已可表示。

4. 必须证明各最小额真正最小，并给出表示。

> Example. With unlimited stamps of values four and seven, find the largest nonrepresentable positive amount.

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离散最值与构造

Method 16 / 方法 16 [LO02-M16] (continued / 续)

> 例题：四与七面额邮票无限，求最大不可表示正金额。

Worked solution

1. Modulo four, zero through three have least represented values $0, 21, 14, 7$: respectively use zero, three, two, one seven-stamps.

2. Adding four-stamps reaches every larger value in the same residue; fewer seven-stamps cannot reach that residue at a lower value.

3. The largest gaps are $-4, 17, 10, 3$, so the largest positive gap is seventeen.

例题解析

1. 模四的零至三余数最小表示额分别为零、二十一、十四、七。

2. 再加四即可覆盖同余类后续所有值；更少七票不能以更小金额得到该余数。

3. 各类最大缺口为负四、十七、十、三，所以最大正缺口为十七。

One large residue minimum is not enough; all other residue classes must also obey the claimed largest gap.

只有一个余数最小额很大还不够，其他所有类也须符合所称最大缺口。

Practice / 配套练习: AIME-2019-II-Q14 (WB p. 515)

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离散最值与构造

Method 17 / 方法 17 [LO02-M17]

Prove the no-gap subset-sum coverage criterion

证明子集和无缺口覆盖准则

Procedure

1. Sort the available positive coin or stamp values and keep the covered interval $[0, S]$.

2. The next value v extends coverage without a gap exactly when $v \leq S + 1$.

3. Then the old interval and its shifted copy overlap or touch, giving all values through $S + v$.

4. Use mandatory denomination counts to prove a lower bound, then exhibit a collection satisfying the criterion and the target total.

通用步骤

1. 将正面额排序，维护已覆盖区间 $[0, S]$。

2. 下一面额 $v \leq S + 1$ 时，且仅此时不会产生新缺口。

3. 原区间与平移区间相接或重叠，覆盖扩展至 $S + v$。

4. 以必须保留的面额数量证明下界，再给出满足准则及目标总量的具体集合。

> Example. Show that stamps $1, 1, 3, 6$ make every amount from zero to eleven, and explain why replacing three by four creates a gap.

> 例题：证明邮票一、一、三、六可拼出零至十一的所有金额，并说明将三改为四为何产生缺口。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 17 / 方法 17 [LO02-M17] (continued / 续)

Worked solution

1. The two ones cover $[0, 2]$. Adding three gives the shifted interval $[3, 5]$, contiguous with it, hence coverage $[0, 5]$.

2. Adding six gives $[6, 11]$, so the full interval through eleven is covered.

3. If three is replaced by four, after the ones the next value exceeds $2 + 1$. The amount three cannot be formed, since all remaining stamps are at least four.

例题解析

1. 两张一覆盖零至二，加入三的平移区间为三至五，与原区间相接。

2. 再加入六，平移部分覆盖六至十一，因此全覆盖。

3. 若三改四，两张一后下一票超过三，金额三无法拼出，因为其他票至少四。

The greedy interval criterion proves all intermediate sums, not merely that the total is large enough.

区间准则证明所有中间和都可达，而不仅是总面额足够。

For denominations $1 < b < c$ with positive counts x, y, z, coverage through the entire collection total is equivalent to $x \geq b - 1$ and $x + yb \geq c - 1$. For a smaller target T, process sorted stamps while the covered endpoint $S < T$: each next stamp must satisfy $v \leq S + 1$; stop once $S \geq T$. Thus a denomination threshold is necessary only if coverage has not already reached T before that denomination is used. This qualification applies to both b and c, as well as requiring total value at least T.

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Method 17 / 方法 17 [LO02-M17] (continued / 续)

三种面额 $1 < b < c$、张数 x, y, z 均正时，覆盖直到全体总额的每个值，充要条件为 $x \geq b - 1, x + yb \geq c - 1$。若仅要求覆盖至较小目标 T，按面额顺序处理邮票：当已覆盖端点 $S < T$ 时，下一票必须满足 $v \leq S + 1$；一旦 $S \geq T$ 即可停止。因此，只有在使用该面额之前尚未覆盖至 T，其门槛才是必要条件；这个限定同时适用于 b、c，并仍要求总额至少为 T。

Practice / 配套练习: AIME-2022-II-Q14 (WB p. 808)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 18 / 方法 18 [LO02-M18]

Use improving coin exchanges to reduce every optimum to a few cases

用改进换币把所有最优解缩成少量情形

Procedure

1. Find local equal-value exchanges that strictly reduce coin count.

2. Every optimum must avoid those patterns, bounding the lower-denomination remainder.

3. This bounds the possible number of largest coins to a short list near the greedy count.

4. Compare every normalized case; greedy is optimal only if none improves it.

通用步骤

1. 寻找同值但严格减少枚数的局部换币。

2. 最优解不能含这些模式，从而限制小面额剩余总值。

3. 由此把大面额枚数限制为贪心数附近的少数候选。

4. 比较全部归一候选，只有无改进时贪心才最优。

> Example. Using coins one, four, six, find the minimum number for amount fourteen and compare with greedy.

> 例题：用一、四、六面额拼十四，求最少枚数，并与贪心比较。

Worked solution

1. An optimum has at most three pennies and two fours, since four pennies exchange for one four and three fours for two sixes.

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离散最值与构造

Method 18 / 方法 18 [LO02-M18] (continued / 续)

2. The non-six remainder is at most eleven, so only two or one sixes need comparison. With two sixes, two pennies give four coins; with one six, two fours give three coins.

3. Thus $6+4+4$ uses the minimum three coins, beating greedy $6+6+1+1$.

例题解析

1. 最优解至多三枚一和两枚四，否则可同值换成更少硬币。

2. 非六剩余至多十一，所以只需比较两枚六或一枚六：前者还需两枚一，共四枚；后者还需两枚四，共三枚。

3. 因此六加四加四为最少三枚，比贪心的四枚更好。

> A normalization proof must cover every optimum, not just a rival example.

> 归一论证必须覆盖所有最优解，不能只给一个竞争例子。

Practice / 配套练习: AIME-2025-II-Q08 (WB p. 805)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 19 / 方法 19 [LO02-M19]

Check degree-weight bounds against graph sign compatibility

检查度数加权界与图符号构造是否相容

Procedure

1. Use a common center to bound each edge difference by the deviations at its two ends.

2. Rearrange deviations against degrees to obtain an upper bound.

3. Equality needs every edge to join compatible low/high labels; test the capacities of those sign classes.

4. If the bound cannot be attained, refine the cases rather than report it as the maximum.

通用步骤

1. 以公共中心将每条边差界为两端偏差之和。

2. 把偏差与度数重排，得到上界。

3. 取等要求每边连接相容高低标号，须检查两符号类的容量。

4. 若不能取等，就细分情形，而不能将上界报成最大值。

> Example. Label a three-leaf star with one through four. Find the greatest sum of edge differences and explain why a naive degree bound can be loose.

> 例题：将一至四标在三叶星图上，求边差总和最大值，并说明简单度数界为何可能偏松。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 19 / 方法 19 [LO02-M19] (continued / 续)

Worked solution

1. At center value $5/2$, deviations are $3/2, 3/2, 1/2, 1/2$. Assigning the largest to degree three gives upper bound seven.

2. If the central vertex has label $c = 1, 2, 3, 4$, the actual three-edge sums are respectively six, four, four, six.

3. Thus the maximum is six. The degree bound seven cannot attain its required edge sign pattern; a bipartite graph alone does not guarantee the needed label capacities.

例题解析

1. 以二分之五为中心，偏差为二分之三、二分之三、二分之一、二分之一，重排度数给出上界七。

2. 但中心顶点分别标一、二、三、四时，真实边差和分别为六、四、四、六。

3. 所以最大为六，七的重排取等符号无法实现；仅有二分图结构还不足以保证标号容量。

An explicit checkerboard construction must allocate each degree class on both colors correctly.

明确棋盘构造还须把每种度数的位置在两色中正确分配。

Practice / 配套练习: AIME-2026-I-Q11 (WB p. 809)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 20 / 方法 20 [LO02-M20]

Split composite counting factors to minimize additive cost

拆分复合计数因子以降低加法成本

Procedure

1. Prove the combinatorial count factors independently as a product of integers $d_i \geq 2$.

2. Express the resource cost of each factor, often $d_i - 1$.

3. If a factor splits as gh, compare the split and unsplit costs exactly.

4. Use prime factorization for the lower bound and construct independent components realizing every prime factor.

通用步骤

1. 先证明组合计数独立分解为若干至少二的整数因子。

2. 表示每因子的资源成本，常为因子减一。

3. 若因子为 gh，精确比较拆分及未拆成本。

4. 以素因数分解给下界，再构造分别实现各素因子的独立部件。

> Example. For $M \geq 2$, prove the minimum of $\sum_i (d_i - 1)$ over all factorizations $M = \prod_i d_i$, $d_i \geq 2$, is $\sum_p v_p(M)(p-1)$.

> 例题：对 $M \geq 2$，证明所有分解 $M = \prod_i d_i, d_i \geq 2$ 中，成本 $\sum_i (d_i - 1)$ 的最小值为 $\sum_p v_p(M)(p-1)$。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 20 / 方法 20 [LO02-M20] (continued / 续)

Worked solution

1. If a factor gh is composite with $g, h \geq 2$, its cost is $gh - 1$. The split cost is $g + h - 2$.

2. The saving is $(g-1)(h-1) > 0$, so no minimum can keep a composite factor.

3. All factors at a minimum are therefore primes. Unique factorization fixes their multiplicities and gives the claimed cost, which the prime-factor list itself attains.

例题解析

1. 复合因子 gh 未拆成本为 $gh - 1$，拆分成本为 $g + h - 2$。

2. 节省量为 $(g-1)(h-1) > 0$，所以最优分解不能保留复合因子。

3. 最优因子全为素数，由唯一分解确定重数，得到公式，素因子列表本身即取等。

The product factorization needs a bijection or unique matching proof before costs may be optimized independently.

独立优化成本之前，计数乘积必须先有双射或唯一匹配证明。

Practice / 配套练习: AIME-2026-II-Q13 (WB p. 718)